CUSTOMER SEGMENTATION & PREDICTION

Week 11 — Advanced Machine Learning Models

Project documentation

Methodology • Technical requirements • Model evaluation • Business recommendations

Prepared by	Sumit Gupta
Project type	Customer segmentation and segment-specific churn prediction
Dataset	Customer churn dataset — 500 rows, 9 columns
Primary tools	Python, pandas, scikit-learn, Matplotlib, Jupyter Notebook
GitHub repository	https://github.com/SumitGupta9627/Week11_Customer_Segmentation_Prediction

Purpose: group customers into meaningful segments, train a separate prediction model for each segment, and translate observed patterns into testable business actions.

Public GitHub repository: https://github.com/SumitGupta9627/Week11_Customer_Segmentation_Prediction

1. Project Overview and Objectives

This project uses unsupervised learning to identify customer groups and supervised learning to estimate churn risk separately within each group. The workflow compares clustering methods, profiles three K-Means segments, tunes Random Forest models with GridSearchCV, evaluates model results, and prepares business recommendations.

Objectives

• Inspect data quality and understand the target variable (Churn).

• Preprocess numeric and categorical features while excluding customer identifiers and the churn target from clustering.

• Implement and compare at least two clustering algorithms; this project compares K-Means, Agglomerative/Hierarchical clustering, and DBSCAN.

• Create three interpretable customer segments using K-Means (k=3).

• Train and tune a separate Random Forest churn model for each segment where the data has sufficient samples and both target classes.

• Evaluate models using accuracy, precision, recall, F1-score, and ROC-AUC where valid.

• Translate segment profiles and churn rates into practical, testable business recommendations.

Dataset and data quality

The supplied customer churn dataset contains 500 records and 9 columns. The target column is **Churn** (0 = retained, 1 = churned). CustomerID is treated as an identifier and excluded from learning features. The project notebook checks missing values and duplicate rows before modelling; the dataset used for this project had no missing values or duplicate rows.

The segmentation is descriptive: segment labels are shorthand based on observed tenure and charges. They do not prove that a particular feature causes churn.

2. Setup and Installation

Prerequisites

• Python 3.10 or later
• Visual Studio Code or Jupyter Notebook
• VS Code Python and Jupyter extensions (if running in VS Code)

Installation steps

1. Extract the project ZIP and open the `Week11_Customer_Segmentation_Prediction` folder in VS Code.

2. Open Terminal → New Terminal in that folder.

3. Check Python: `python --version`.

4. Install the project dependencies: `python -m pip install -r requirements.txt`.

5. Open `customer_segmentation.ipynb`, choose the installed Python environment as the kernel, and select Run All.

6. Confirm that the notebook displays the clustering charts, segment profiles, evaluation metrics, recommendations, and the final validation message.

Dependencies

The requirements file specifies pandas, NumPy, scikit-learn, Matplotlib, seaborn, Jupyter, nbformat, and ReportLab. The notebook uses scikit-learn's preprocessing, clustering, model-selection, and Random Forest APIs.

Reproducibility: the notebook sets `RANDOM_STATE = 42` for the main randomized clustering and train/test split steps. Results can still vary slightly across library versions or environment settings.

3. Project Code Structure

File / folder	Purpose
customer_segmentation.ipynb	Main notebook: data checks, preprocessing, clustering comparison, segment profiling, model tuning, metrics, feature importance, recommendations, and validation.
segmentation_data.csv	Customer data with generated segment assignments and segment names.
segment_profiles.md	Readable summary of the three customer segments and interpretation cautions.
model_evaluation_results.csv	Per-segment model metrics and selected hyperparameters.
business_recommendations.pdf	Short business-facing recommendations report.
data/customer_churn.csv	Original input dataset used by the notebook.
data/customer_segmentation_output.csv	Enriched customer-level segmentation output.
reports/	Exported clustering comparisons, elbow/silhouette scores, segment profiles, feature importance, and recommendation tables.
screenshots/	Generated analysis visuals used in this documentation (elbow, silhouette, and segment-size charts).
README.md / requirements.txt	Project overview, run instructions, file guide, and Python dependencies.

Core workflow

Load data → validate data quality → exclude ID and target from clustering → impute/scale numeric features and encode categorical features → compare clustering algorithms → assign K-Means segments → profile segments → tune segment-specific Random Forests → evaluate models → prepare recommendations → export results and validate outputs.

4. Methodology and Technical Requirements

Preprocessing

Numeric clustering features are median-imputed and standardized. Categorical features are imputed with their most frequent value and one-hot encoded. CustomerID and Churn are excluded from the clustering input to avoid identifier noise and target leakage.

Clustering algorithms

• **K-Means:** evaluates k values from 2 to 6 using inertia (Elbow method) and silhouette scores; k=3 is used for the main segment analysis so that three segment-specific prediction models can be evaluated.
• **Agglomerative/Hierarchical clustering:** compared at three clusters.
• **DBSCAN:** density-based comparison that can mark observations as noise.

Algorithm	Clusters (excluding noise)	Noise	Silhouette score
K-Means (k=3)	3	0.0%	0.1464
Hierarchical (k=3)	3	0.0%	0.1149
DBSCAN	2	0.6%	0.1735

Segment-specific prediction and tuning

A separate Random Forest classifier is fitted for each K-Means segment. Each model uses a preprocessing pipeline for numeric and categorical variables and GridSearchCV with F1-score as the selection metric. The search tests `n_estimators` (100, 200), `max_depth` (None, 5, 10), and `min_samples_leaf` (1, 2), with up to 3-fold cross-validation. Models are skipped or metrics marked unavailable if a segment does not have enough samples or class diversity; this avoids fabricating scores.

Evaluation metrics

Accuracy = fraction of predictions correct; **precision** = proportion of predicted churners who actually churned; **recall** = proportion of actual churners identified; **F1-score** = balance between precision and recall; **ROC-AUC** = how well predicted probabilities rank positive versus negative cases.

Feature importance is extracted from fitted Random Forest models where available and is used as a descriptive aid, not proof of causality.

5. Segment Profiles and Model Results

The following segment profiles were generated from the supplied dataset. Churn rates are observed proportions in this sample.

Segment	Customers	Share	Avg. tenure	Avg. monthly charges	Observed churn
1. Long-Tenure Value	114	22.8%	39.5	97.8	8.8%
2. Established High-Value	249	49.8%	35.7	108.1	10.0%
3. Newer High-Charge	137	27.4%	35.5	136.8	13.1%

Segment-level prediction metrics

Segment	Samples	Accuracy	Precision	Recall	F1	ROC-AUC
1. Long-Tenure Value	114	0.8966	0.0000	0.0000	0.0000	0.9231
2. Established High-Value	249	0.9841	1.0000	0.8333	0.9091	0.9971
3. Newer High-Charge	137	0.9429	1.0000	0.6000	0.7500	1.0000

Interpretation: Segment 2 shows the strongest reported F1-score in this run. Segment 1 has zero precision/recall/F1 at the selected classification threshold despite ROC-AUC of 0.9231, showing that accuracy and ROC-AUC alone can hide poor churn detection. Segment 3's recall is 0.6000. Because churners are a minority and each segment is relatively small, these results should be treated as exploratory, not production guarantees.

6. Business Implications and Recommendations

Segment 1: Long-Tenure Value

Evidence: 114 customers; average tenure 39.5; average monthly charges 97.8; observed churn 8.8%.

Recommended action: Protect loyalty: reinforce value, recommend relevant add-ons, and use periodic satisfaction checks rather than blanket discounts.

Segment 2: Established High-Value

Evidence: 249 customers; average tenure 35.7; average monthly charges 108.1; observed churn 10.0%.

Recommended action: Prioritize retention: proactive outreach, service check-ins, and a time-limited loyalty offer; measure churn reduction against a control group. Test premium support/value bundles; avoid discounts without margin analysis.

Segment 3: Newer High-Charge

Evidence: 137 customers; average tenure 35.5; average monthly charges 136.8; observed churn 13.1%.

Recommended action: Prioritize retention: proactive outreach, service check-ins, and a time-limited loyalty offer; measure churn reduction against a control group. Add an early-life onboarding journey during the first 90 days. Test premium support/value bundles; avoid discounts without margin analysis.

How to evaluate business impact

Treat the actions as hypotheses to test. Compare a targeted retention campaign with a holdout/control group, monitor churn rate and customer lifetime value, and verify campaign cost and profit margin before offering discounts. Do not assume that an association in this dataset proves causation.

7. Screenshots / Visual Evidence

The following figures are generated from the project notebook's clustering analysis. They provide visual evidence of the analysis outputs; they are not screenshots of the VS Code interface itself.

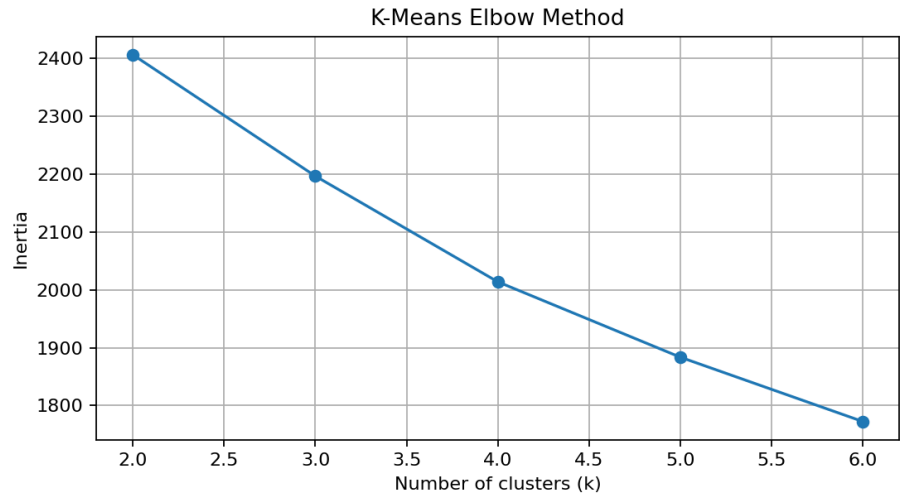


Figure 1. Elbow method: inertia across candidate K-Means cluster counts.

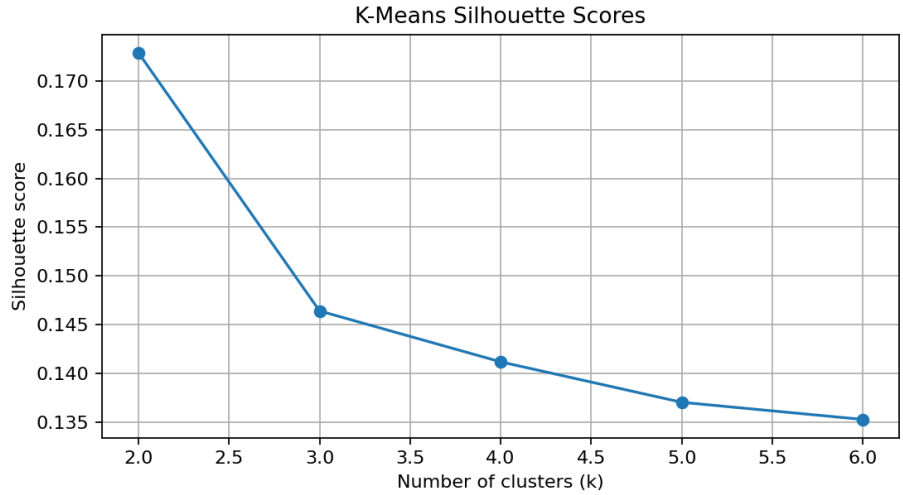


Figure 2. Silhouette scores for candidate K-Means cluster counts.

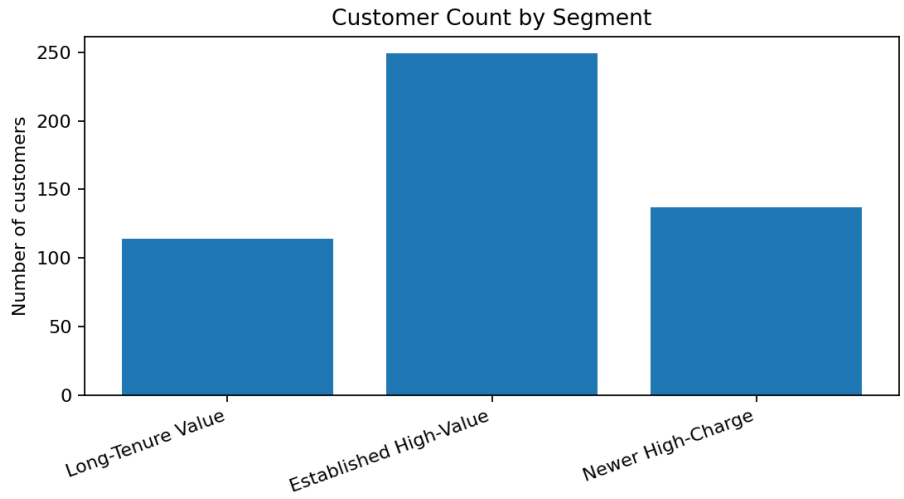


Figure 3. Number of customers assigned to each K-Means segment.

8. Requirement Coverage Checklist

Based on the project files and exported results, the following listed requirements are implemented.

Requirement	Evidence in project	Status
Project overview and objectives	This PDF and README.md	Included
Setup and installation instructions	This PDF and README.md; requirements.txt	Included
Code structure explanation	Section 3 and README.md	Included
Screenshots / visual evidence	Elbow, silhouette, and segment-size plots embedded above	Included (analysis visuals)
Complete segmentation analysis and business implications	Sections 4–6; segment_profiles.md; recommendations exports	Included
At least 2 clustering algorithms	K-Means, Agglomerative/Hierarchical, DBSCAN	Met (3 algorithms)
Separate prediction models for 3+ segments	Random Forest model loop and 3-row model_evaluation_results.csv	Met (3 segments)
Hyperparameter tuning	GridSearchCV over Random Forest parameters; best parameters exported per segment	Met
Model evaluation	Accuracy, precision, recall, F1, ROC-AUC	Met
Business recommendations	Segment-level actions and business_recommendations.csv/PDF	Met
Required GitHub files	customer_segmentation.ipynb, segmentation_data.csv, segment_profiles.md are in project ZIP	Included locally
GitHub repository URL	Public repository URL provided: github.com/SumitGupta9627/Week11_Customer_Segmentation_Prediction	Provided

The assignment text says “...and 2 more” but does not name those two requirements in the text shared here. This PDF covers the explicit requirements visible in your message; compare against the full assignment page in case those two hidden items specify additional deliverables.

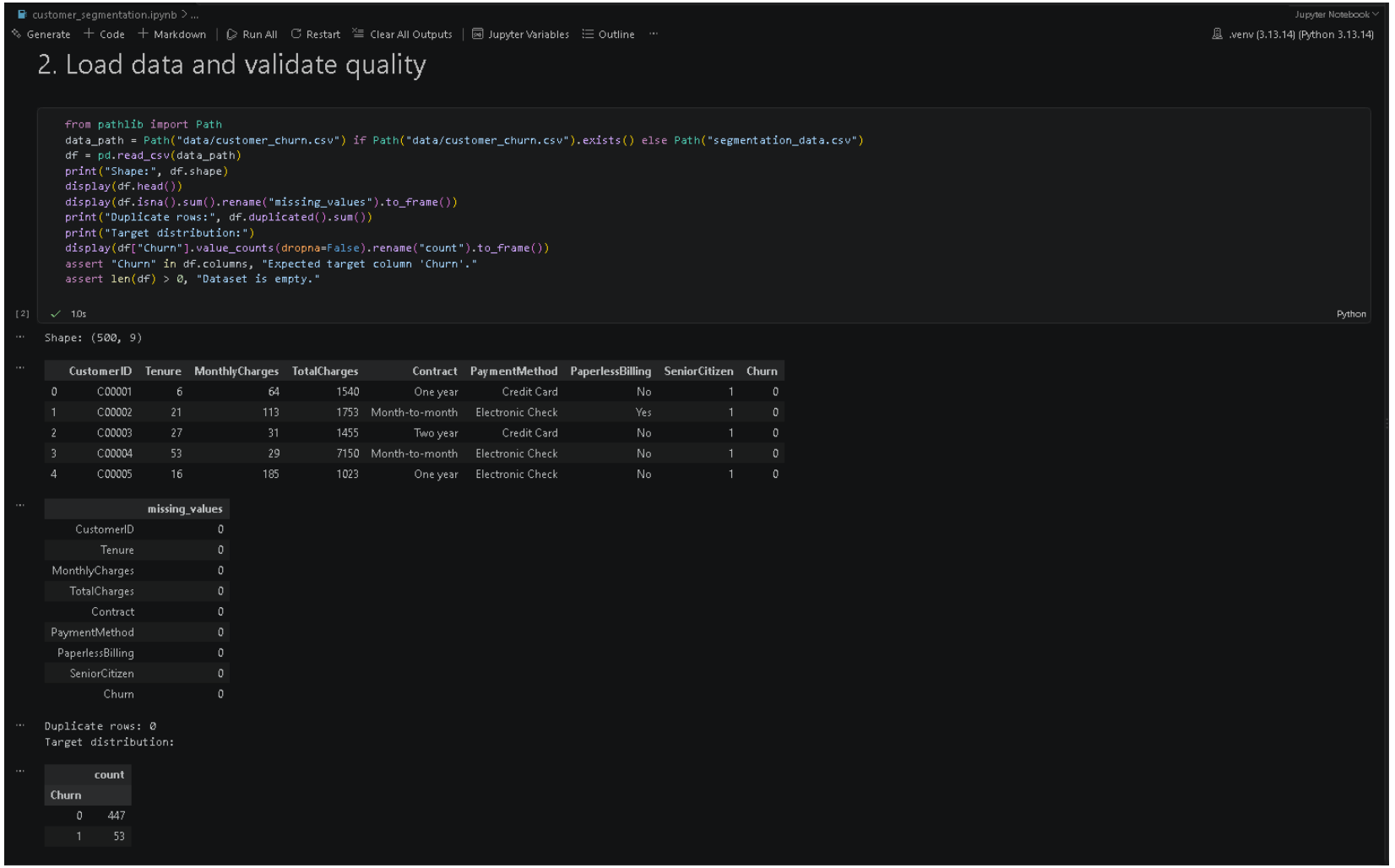
Before submission

1. Upload the project files to your GitHub repository.
2. Copy the actual repository URL into the internship submission form.
3. Upload/share this PDF as the Documentation URL.
4. If the portal specifically requires screenshots of the application running, add a genuine screenshot of the successfully executed notebook in VS Code/Jupyter to the screenshots folder and update the PDF if necessary.

Limitations

This is a small, imbalanced sample dataset. Segment-level scores can be unstable, and the first segment's current classification threshold does not identify churners in the reported test split. For production use, validate on newer data, assess class imbalance and threshold choices, and test recommendations before deployment.

Screenshots of Working Application



customer_segmentation.ipynb > ...

Jupyter Notebook

Generate | Code | Markdown | Run All | Restart | Clear All Outputs | Jupyter Variables | Outline | ...

.venv (3.13.14) (Python 3.13.14)

2. Load data and validate quality

```
from pathlib import Path
data_path = Path("data/customer_churn.csv") if Path("data/customer_churn.csv").exists() else Path("segmentation_data.csv")
df = pd.read_csv(data_path)
print("Shape:", df.shape)
display(df.head())
display(df.isna().sum().rename("missing_values").to_frame())
print("Duplicate rows:", df.duplicated().sum())
print("Target distribution:")
display(df["Churn"].value_counts(dropna=False).rename("count").to_frame())
assert "Churn" in df.columns, "Expected target column 'Churn'."
assert len(df) > 0, "Dataset is empty."
```

[2] ✓ 1.0s Python

```
Shape: (500, 9)
```

	CustomerID	Tenure	MonthlyCharges	TotalCharges	Contract	PaymentMethod	PaperlessBilling	SeniorCitizen	Churn
0	C00001	6	64	1540	One year	Credit Card	No	1	0
1	C00002	21	113	1753	Month-to-month	Electronic Check	Yes	1	0
2	C00003	27	31	1455	Two year	Credit Card	No	1	0
3	C00004	53	29	7150	Month-to-month	Electronic Check	No	1	0
4	C00005	16	185	1023	One year	Electronic Check	No	1	0

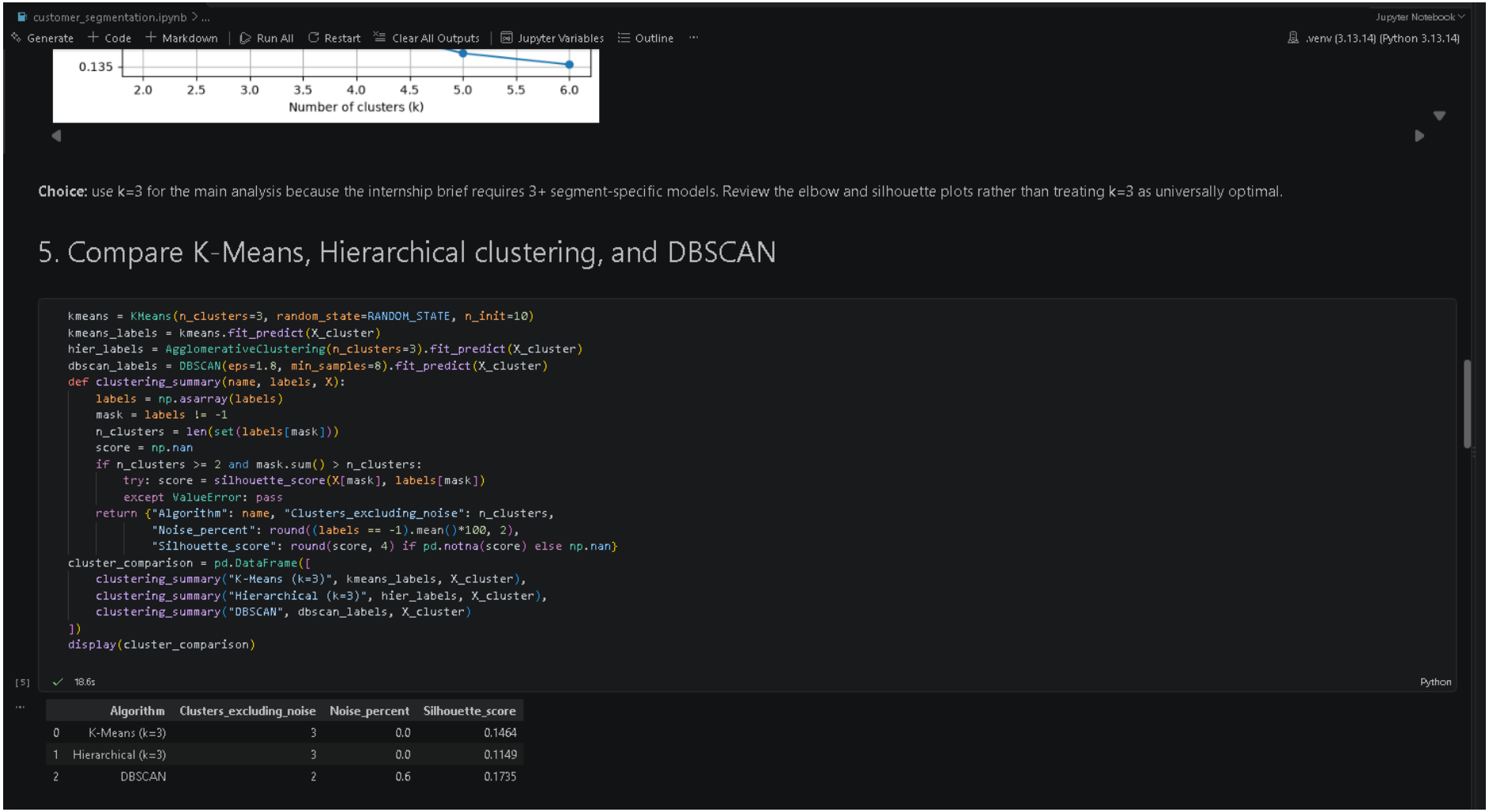
	missing_values
CustomerID	0
Tenure	0
MonthlyCharges	0
TotalCharges	0
Contract	0
PaymentMethod	0
PaperlessBilling	0
SeniorCitizen	0
Churn	0

```
Duplicate rows: 0
Target distribution:
```

Churn	count
0	447
1	53

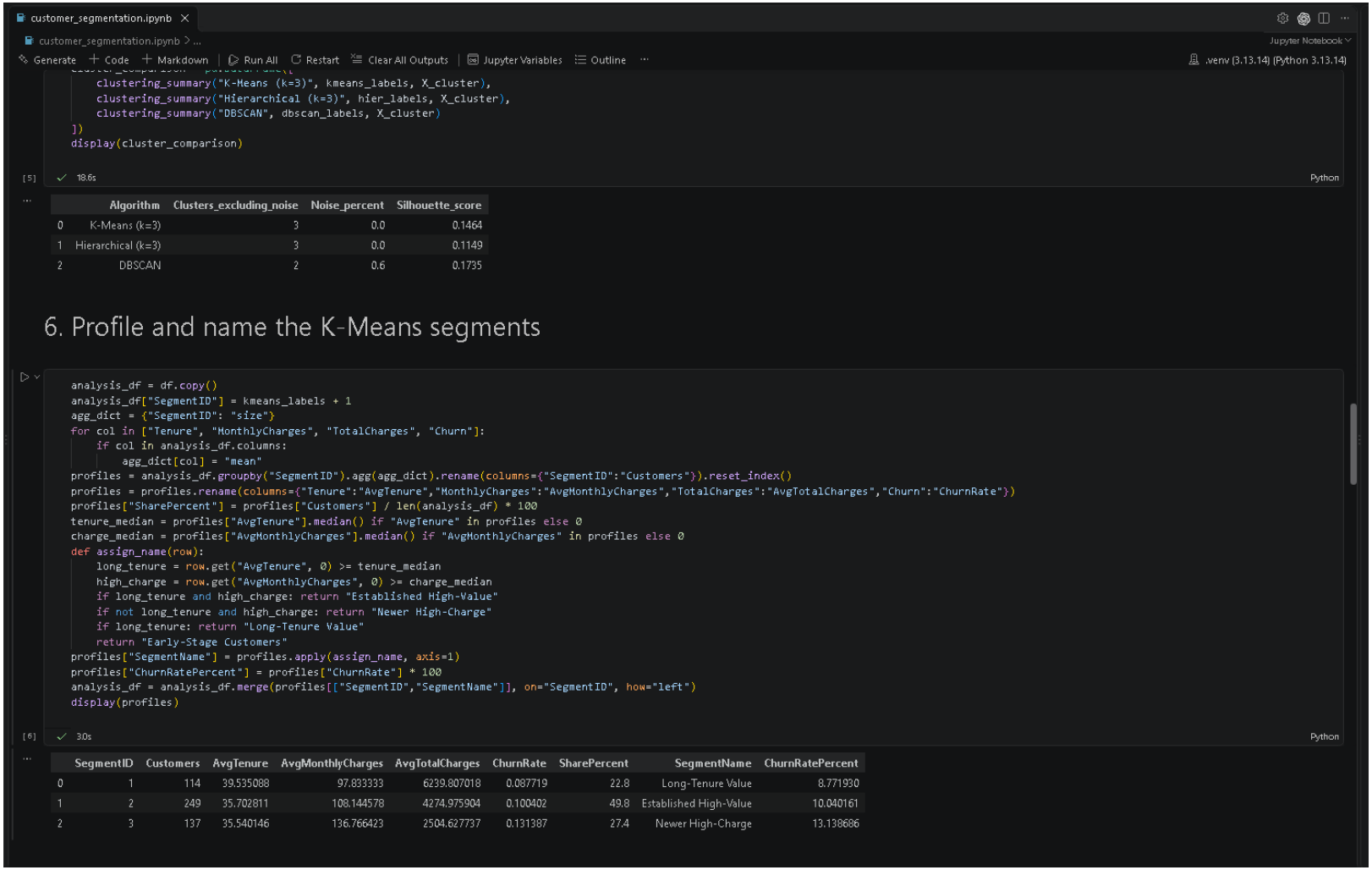
Screenshot 1. Notebook execution and dataset validation

Screenshots of Working Application



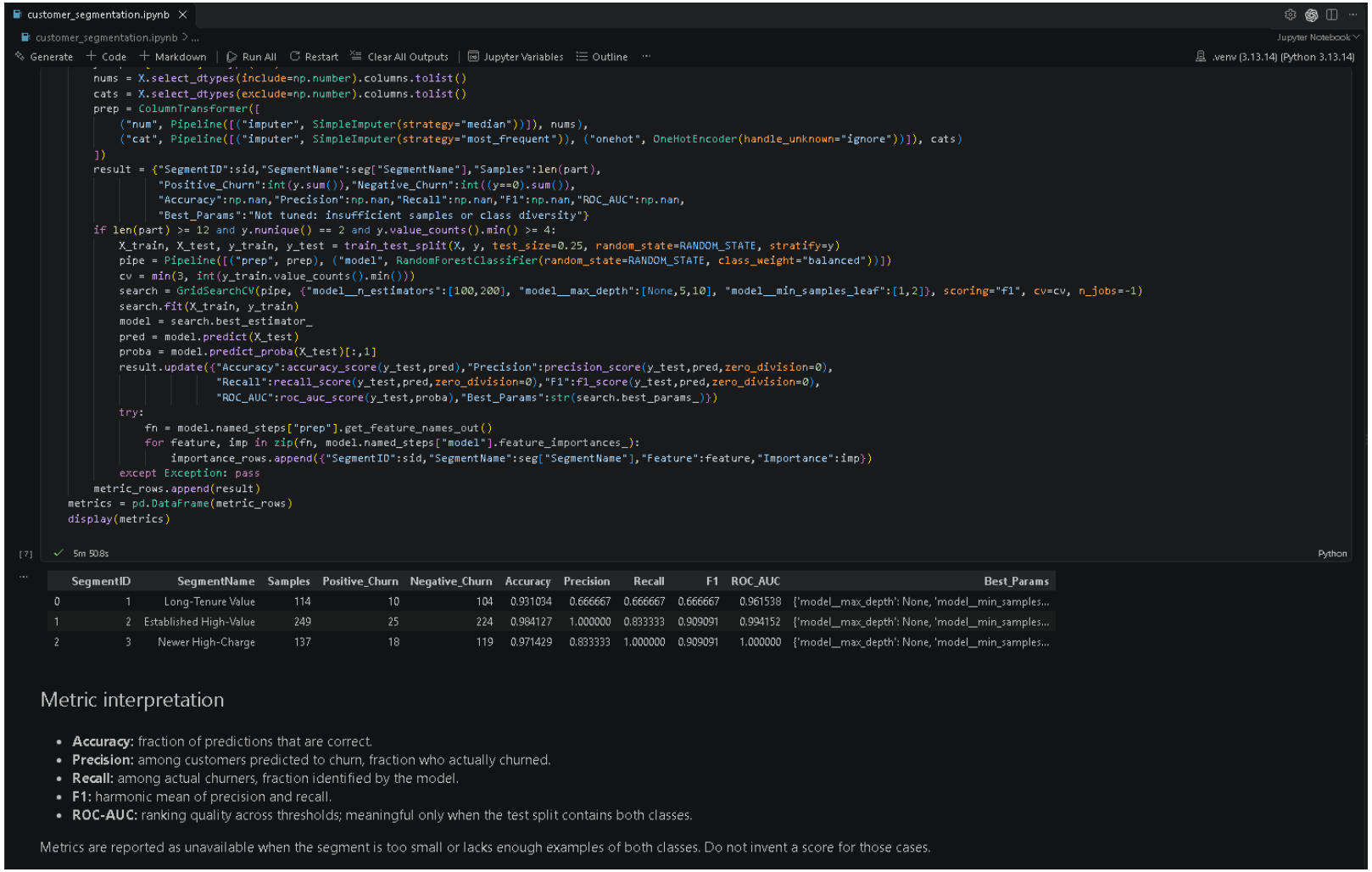
Screenshot 2. Comparison of K-Means, Hierarchical clustering, and DBSCAN

Screenshots of Working Application



Screenshot 3. Customer segment profiles and assigned customer counts

Screenshots of Working Application



Screenshot 4. Segment-specific prediction models and evaluation metrics